LINEAR INEQUALITIES

1. Introduction

- An inequality shows a relationship between two expressions that are not necessarily equal.
- Linear inequality involves polynomials of degree 1 (i.e., linear expressions).
- We use $<, >, \leq, \geq, \neq$ to compare two quantities.

Examples : $2x + 3 > 7$, $3x - 5 \leq 4$, $x + 2 < 0$

$5x + 1 \geq 3$, $x - 4 \neq 0$

Symbols

Symbol	Read as	Meaning
$<$	less than	LHS is smaller than RHS
$>$	greater than	LHS is greater than RHS
$\leq$	less than or equal to	LHS is smaller than or equal to RHS
$\geq$	greater than or equal to	LHS is greater than or equal to RHS
$\neq$	not equal to	LHS is not equal to RHS

2. Symbols of Inequalities

There are two types of inequalities.

(i) Strict inequalities : $<$, $>$

(ii) Non-strict inequalities : $\leq$, $\geq$

3. Rules for Solving Linear Inequalities

① If we add (or subtract) the same number (or expression) to (or from) both sides, the direction of inequality remains the same.

② If we multiply (or divide) both sides by a positive number, the direction of inequality remains the same.

③ If we multiply (or divide) both sides by a negative number, the direction of inequality is reversed.

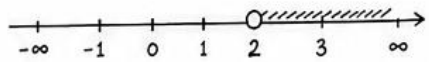
> * Always remember to reverse the sign when multiplying or dividing by a negative number.

4. Solution on the Number Line

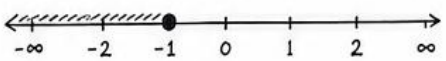
- Open circle (o) → for $<$ or $>$
- Closed circle (●) → for $\leq$ or $\geq$
- Shade the part which represents the solution set.

Examples :

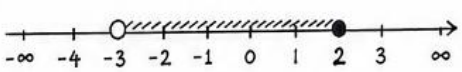
(i) $x > 2$ →



(ii) $x \leq -1$ →



(iii) $-3 < x \leq 2$ →



5. Linear Inequalities in One Variable

General form : $ax + b < 0$, $ax + b > 0$,

$ax + b \leq 0$, $ax + b \geq 0$ $(a \neq 0)$

Steps to solve :

① Remove brackets.

② Collect the term containing x on one side and constant on the other side.

③ Simplify.

④ Divide (or multiply) by the coefficient of x.

⑤ If we divide (or multiply) by a negative number, reverse the inequality sign.

Examples :

(i) $3x - 5 > 7$

$3x > 12$

$x > 4$



(ii) $-2x + 1 \leq 7$

$-2x \leq 6$

$x \geq -3$



(iii) $7 - 5x < -3$

$-5x < -10$

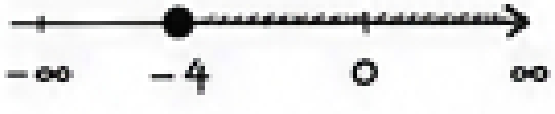
$x > 2$



(iv) $4 + 3x \geq -8$

$3x \geq -12$

$x \geq -4$



6. Linear Inequalities in Two Variables

- General form : $ax + by + c < 0$ or $ax + by + c > 0$

  (a, b not both zero)

- The equation $ax + by + c = 0$ represents a straight line in the xy-plane.
- The inequality represents one of the two half-planes determined by this line.
- Every point on the line satisfies $ax + by + c = 0$.
- Points on one side satisfy $ax + by + c > 0$ and on the other side satisfy $ax + by + c < 0$.
- To find the region, choose any test point not on the line and check the sign.

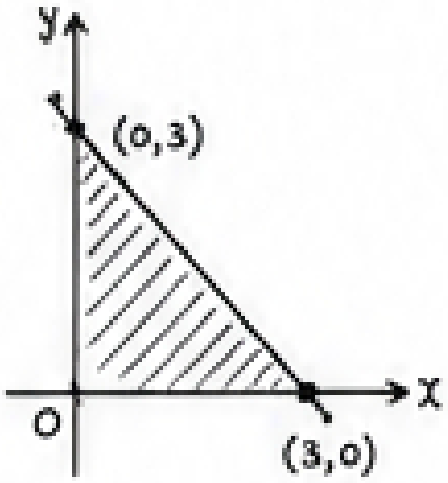
Example : $x + y - 3 > 0$

Line : $x + y - 3 = 0 \rightarrow x + y = 3$

Intercepts :

Put $x = 0 \rightarrow y = 3$ point $(0, 3)$

Put $y = 0 \rightarrow x = 3$ point $(3, 0)$



Test point : $(0, 0)$

$0 + 0 - 3 = -3 < 0$ (not satisfying)

Hence, the solution region is on the side not containing $(0, 0)$.

7. Graphical Representation & Feasible Region

- The solution set of a linear inequality in two variables is the set of all points in one of the half-planes (including or excluding the boundary line).
- The boundary line is drawn using the equation $ax + by + c = 0$.
- If the inequality is strict ($<$ or $>$), the boundary line is shown by a dotted line.
- If the inequality is non-strict ($\leq$ or $\geq$), the boundary line is shown by a solid line.
- The solution set (feasible region) is the region that satisfies all the given inequalities (the common region).

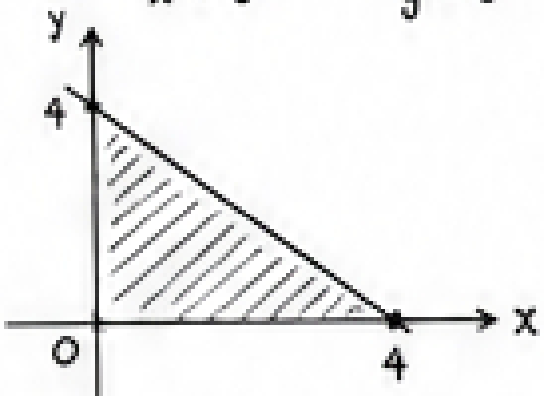
Steps :

① Draw the boundary line for each inequality.
② Take a test point (usually (0,0)) not on the line and check the inequality.
③ Shade the region that satisfies it.
④ The common shaded region is the feasible region.

Example : $x + y \leq 4$, $x \geq 0$, $y \geq 0$

Lines : $x + y = 4$ (intercepts (4,0) and (0,4)

$x = 0$ $y = 0$



Shaded region (in first quadrant) is the feasible region.

5. Linear Inequalities in One Variable

General form : $ax + b < 0$, $ax + b > 0$,

$ax + b \leq 0$, $ax + b \geq 0$ $\quad (a \neq 0)$

Steps to solve :

1. Isolate the term containing x.
2. Bring the constant term to the other side.
3. Divide (or multiply) by the coefficient of x.
4. If we divide by a negative number, reverse the inequality sign.

Examples :

(i) $3x - 5 > 7$

$3x > 12$

$x > 4$



(ii) $-2x + 1 \leq 7$

$-2x \leq 6$

$x \geq -3$



(iii) $5 - 3x < -10$

$-3x < -15$

$x > 5$



(iv) $\frac{x+2}{3} \geq 2$

$x + 2 \geq 6$

$x \geq 4$



Important Points :

- Inequalities have infinite number of solutions.
- Solution set is always an interval.
- We do NOT get a single value in solution.

Types of Intervals in R :

(i) $(-\infty, a)$	(ii) $(-\infty, a]$	(iii) (a, ∞)
(iv) $[a, \infty)$	(v) (a, b)	(vi) $[a, b]$